CMP020X305S: BSc Cyber Security

Title of coursework: Web Application Vulnerabilities

About this portfolio lab

This portfolio lab is problem centred. Therefore, you will need to research and take responsibility for solving any technical challenges that you encounter. You are encouraged to work collaboratively with other students in your cohort. However, the work that you submit must be your own, and any sources of assistance, etc., must be acknowledged through referencing or, where appropriate, through code comments.

Getting Started

1. Establish a network between your Kali Linux virtual machine and your OWASP Juice Shop virtual machine.

   KALI
   - The username to login is `kali` and the password is `kali`
   - Give this machine an IP address of `192.168.123.100`

   OWASP Juice Shop
   - You can download a VirtualBox Juice Shop VM from the UBS provided during the lab session.
   - The username to login is `student`, and the password is `Student1`
   - This virtual machine already has an IP address of `192.168.123.111`

2. Ensure that your Kali virtual machine and your OWASP Juice Shop virtual machine can ping each other.

3. Login to the OWASP Juice Shop virtual machine.

4. Change the directory to the `juice-shop` folder by typing...

   ```
   cd juice-shop
   ```

5. Start the webserver by typing...

   ```
   sudo npm start
   ```

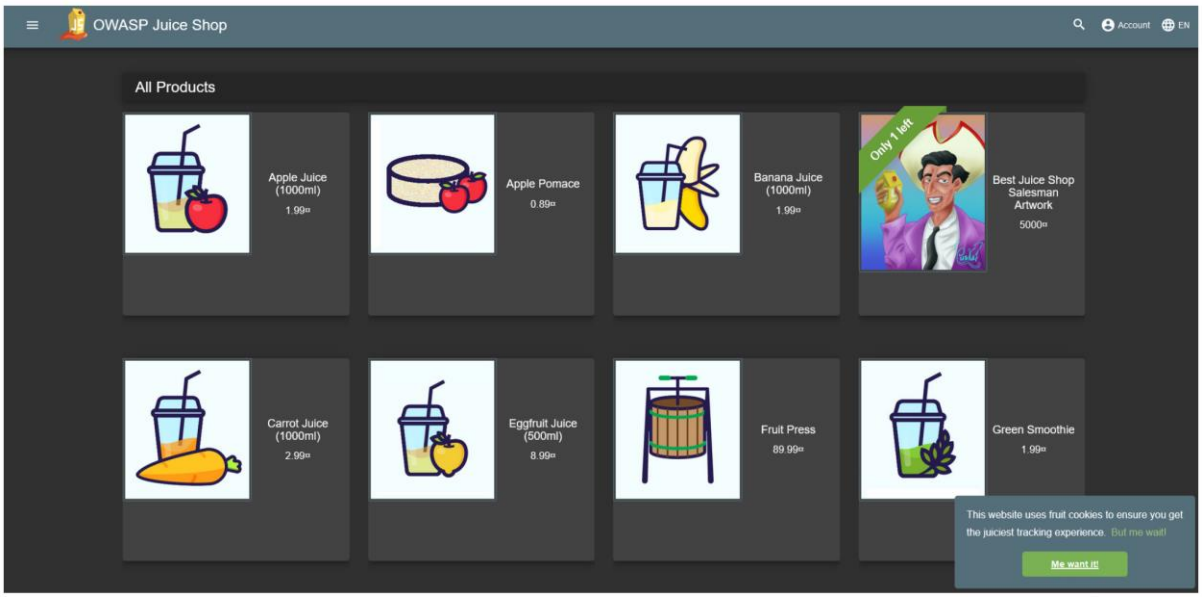
   The server should start and will show the following status...

```
student@juiceshop:~$ cd juice-shop
student@juiceshop:~/juice-shop$ sudo npm start
[sudo] password for student:

> juice-shop@14.5.1 start
> node build/app

info: All dependencies in ./package.json are satisfied (OK)
info: Chatbot training data botDefaultTrainingData.json validated (OK)
info: Detected Node.js version v16.19.1 (OK)
info: Detected OS linux (OK)
info: Detected CPU x64 (OK)
info: Configuration default validated (OK)
info: Entity models 19 of 19 are initialized (OK)
info: Required file server.js is present (OK)
info: Required file index.html is present (OK)
info: Required file tutorial.js is present (OK)
info: Required file runtime.js is present (OK)
info: Required file vendor.js is present (OK)
info: Required file styles.css is present (OK)
info: Required file main.js is present (OK)
info: Required file polyfills.js is present (OK)
info: Port 3000 is available (OK)
info: Server listening on port 3000
```

6. From your Kali virtual machine, open a browser and visit...

```
192.168.123.111:3000
```

The following page should load:



TASK 01: Carefully read, review and explore information about OWASP Juice Shop. Feel free to use any resources that you like. Here are a few links to get you started:

- https://owasp.org/www-project-juice-shop/
- https://pwning.owasp-juice.shop/
- https://akash-pawar.github.io/owasp-juice-shop/

Deliverables

OWASP Juice Shop has many vulnerabilities. Having read and researched OWASP Juice Shop, you must attempt up to 10 Juice Shop challenges and provide evidence of completion as screen shots from Juice Shop, as well as an in-lab demonstration.

For each challenge that you complete, you will need to include:

1. The name of the challenge.
2. The difficulty star rating of the challenge.
3. The evidence in the form of a screenshot

End of Portfolio :-)